

# RIBBON-shape-options-addendum

## RIBBON — Shape Options Addendum: Ring & Hex

Rev D companion — extends the racetrack flagship with two geometries

The height system (560–1100 mm), base requirements B1–B10, colorways, and trough/insert logic are unchanged; only the plan geometry varies. Companion drawing: RIBBON-shape-options-plan-view.svg.

### 1 · RIBBON Ring (circular)

- **Geometry:**  $\varnothing \approx 2.95$  m annulus; **ring depth 500 mm**; center core  $\varnothing \approx 1.1$  m (the unreachable middle becomes the feature: planter, ice well, or supply hub). 14 positions on  $\sim 9.2$  m of circumference.
- **Wave:** rises and falls **twice** around the circle (two mirrored gradients), so the surface returns to its start height — a continuous helical rim with no step.
- **Accessibility:** a **flattened chord** on the low arc creates the knee-clear 740 mm bays (a pure circle can't hold a wheelchair square-on); B5 clearances preserved.
- **Trough:** becomes a concentric **ring trough** mid-annulus; inserts are arc segments.
- **Trade-offs (honest):** compound-curved thermoforming is the hardest of the three shapes (higher mold cost, more scrap); no straight module joints, so **Ring does not fold** and modularizes worst; everyone faces inward — the most social geometry, the least workable for laptops-in-rows.
- **Best for:** lobbies, museums, story-circles, tasting rooms — occasions, not workstations.

### 2 · RIBBON Hex (hexagonal)

- **Geometry:** 2.65 m across flats; six  $\sim 1.53$  m edges; positions 2-2-3-2-2-3 = **14**. Center hex well (or turntable) solves the deep middle.
- **Heights: one plateau per edge** — six clean height zones (toddler → bar), with short ramped corners handling every transition. The simplest possible statement of the multi-height idea.
- **Accessibility:** one full edge (plus part of a second) at 740 mm knee-clear — flat edges make Hex the easiest shape to roll up to.
- **Engineering win:** all-straight segments — **no compound curves**, simplest thermoform (or even flat stock + ramped corner pieces), cleanest module kit, and the natural **Fold** candidate (two hinge lines across the width fold it like a pie case).
- **Scaling story:** hexes **tessellate** — venues can tile two or three RIBBON Hexes into honeycomb clusters, which no other shape offers. This turns a table into a floorplan system.
- **Best for:** schools and libraries (clusters!), makerspaces, cafeterias, coworking.

### 3 · The shape family, summarized

|              | Racetrack (flagship)    | Ring                            | Hex                        |
|--------------|-------------------------|---------------------------------|----------------------------|
| Footprint    | $3.9 \times 1.2$ m      | $\varnothing 2.95$ m            | 2.65 m A/F                 |
| Wave         | Continuous, 2 long runs | Helical, rises/falls $\times 2$ | 6 plateaus, ramped corners |
| ↳ bays       | 4 (both ends)           | 2 (flattened chord)             | 2–3 (flat edges)           |
| Fold variant | Yes (tri-fold)          | No                              | Yes — best folder          |

|                       | Racetrack (flagship) | Ring                     | Hex                 |
|-----------------------|----------------------|--------------------------|---------------------|
| Thermoform difficulty | Medium               | <b>Hardest</b>           | <b>Easiest</b>      |
| Modularity / kit fit  | Good                 | Poor                     | <b>Best</b>         |
| Sociability           | Linear + ends        | <b>Max (all face in)</b> | Clustered           |
| Unique power          | Length & flow        | The gathering circle     | <b>Tessellation</b> |
| Price posture         | Baseline             | Premium (+mold, +scrap)  | At/below baseline   |

#### 4 · Ripple effects across the project

- **Configurator:** shape becomes the *first* dimension: **Racetrack · Ring · Hex** → then variant (One/Fold, where offered) → height mix → colorway → inserts. Still one parametric file per order.
- **Build spec:** B1-B10 apply per shape; Ring adds a chord-flat requirement; Hex adds corner-ramp and tessellation-edge specs.
- **Crossover model:** Hex slightly *lowers* materials/tooling vs. baseline; Ring raises them (~+10–20% at pilot volume, mold-driven). Test via the Materials and Tooling yellow cells.
- **Roadmap:** Racetrack stays the Phase 0–1 flagship (deepest validation value). **Hex prototypes in Phase 1–2** — it's the volume/education play and the LFAM-friendliest geometry. **Ring waits for Phase 2+** as the premium statement piece once mold economics are proven.
- **Pilots:** the education pilot should see a **Hex cluster** (two tessellated units) — that tests the shape *and* the floorplan-system pitch in one site.
- **Deck & one-pager:** shape family added; deck gains a shape-family slide.
- **Market angle:** Hex clusters open a per-site multi-unit sale (2–4 tables/room), materially raising average order value in education — worth reflecting in the financial refresh when real quotes land.

#### 6 · Rev D.2 — the Fixed Focal Section becomes a family-wide design law

Every RIBBON shape now carries a **fixed focal section**: one permanent, structural, service-carrying element that is the visual and functional heart of the table, **typed at order** (Planted · Service · Ice/beverage · Solid) and never swapped.

- **Ring:** the center core (as in Rev D.1) — mast + radial arms.
- **Hex:** the center hub (as in Rev D.1) — mast-supplemented.
- **Racetrack (new):** the **middle ~700 mm of the central trough** becomes the fixed focal section, seated directly over the spine at mid-span between the two pedestals — the strongest point of the structure. Services (power riser, irrigation, or drain) route through the spine to it. The trough segments flanking it on either side remain fully swappable, as do all activity wells.

**Why one law across the family:** - **Identity:** every RIBBON has a recognizable heart — the thing you see first and gather around — which gives the brand a signature across three very different geometries. - **Structure & services:** the focal section always sits on the strongest load path (mast centroid or spine mid-span), so it is the natural riser for power and water on every shape. - **Clear split of permanence:** *focal section = committed at order; perimeter inserts + wells = the swappable mode layer*. One sentence explains the whole product logic to a buyer.

**Spec impact:** B11 is generalized from “center mast” to “**fixed focal section**” and applies to all shapes: ≥200 kg service load, internal-only routing, no exposed conductors or water lines within child reach, anti-tip (B3) verified with the focal load in place. Racetrack’s focal section additionally may not reduce the swappable trough run below 1.0 m per side.